STAT 486 Project Plan

Due: Tuesday, October 27, 2026, at 11:59 PM Mountain Time
Team: [Names]
Repository: [GitHub URL]

Use these prompts to plan your investigation. Keep the answers focused; revise the plan when evidence or instructor feedback changes your direction. The Oct. 27 checkpoint still requires a new, dated project-plan entry in the root `README.md` of your team's GitHub repository. Link this document there if you use it; this document does not replace that entry.

Project question and purpose

- What specific question do you want to investigate?
- Why does this question interest your team, and what might someone learn from answering it?
- What claim or expectation could your evidence support, challenge, or leave uncertain?
- What does one observation represent in the data you expect to use?

Chosen route

Choose one route and explain why it fits your question and available time.

- ☐ **A. Develop something new** in an ML method or system
- ☐ **B. Kaggle competition** with two meaningfully different pipelines
- ☐ **C. Build a dataset** and use it in a reproducible ML pipeline

Why this route: [How does the route support the question and the team's intended contribution?]

Data source and access

- What data will you use? Give the dataset, competition, API, or source name and a direct link if available.
- What does each row or record represent, and which fields are relevant to the question?
- How will the team obtain the data? Note access requirements, collection dates or versions, and any expected size or compute needs.

- What license, competition rules, privacy, ethical, or other use limits apply? Could any data be restricted or identifying?
- What data quality issues might affect the investigation (for example, missingness, duplicates, labels, coverage, or changes over time)?

Route-specific plan

Complete the subsection for your chosen route. For a different route, remove the unused subsections.

YOU DO NOT NEED TO ANSWER EVERY SINGLE QUESTION; THESE ARE TO PROVIDE IDEAS OF CONSIDERATIONS TO THINK ABOUT.

Route A: New method or system component

- What simple reference method will you implement first?
- What is the one team-designed change you want to test, and what purpose does it serve?
- How will you isolate the effect of that change under comparable evaluation conditions?
- What small or synthetic case can help check that the component works as intended?
- What result or setting might show that the change does not help?

Route B: Kaggle competition

- Which competition are you considering, and has the instructor approved it?
- Do its rules allow the team's planned data and code use? Are data access and compute feasible by Dec. 10?
- What will pipeline 1 do, from competition data through preparation, local validation, training, and inference?
- What substantive design decision will differ in pipeline 2 (not only a seed or minor hyperparameter)? Why investigate that change?
- What local validation approach and metric will you use? How will you guard against leakage?
- How will you record each pipeline's run command, configuration, commit, valid Kaggle submission, local result, and Kaggle result?

Route C: New dataset and ML pipeline

- What useful dataset will you create, and what question will the ML analysis investigate?
- What source or sources will you combine or collect? Are collection and redistribution permitted?
- What does one observation represent, and what will the dataset schema contain?
- How will you collect, integrate, clean, or label the data? Which scripts will reproduce these steps?

- How will you record source locations, access or license terms, collection dates or versions, and transformations?
- What quality checks will you run (such as join coverage, duplicates, missingness, label consistency, or source changes)?
- Can you preserve a permitted snapshot or small fixture? If raw data cannot be shared, how can an authorized reader rebuild or verify the cleaned data?
- What reference ML method will use the cleaned dataset, and what evidence will show both dataset fitness and ML usefulness?

Initial reference and proposed contribution

- What is the simplest credible baseline, reference method, or comparison for your question?
- What will your team design, build, collect, or investigate beyond that reference?
- Which choices, code, data, papers, notebooks, packages, or pretrained models will come from others, and how will you credit them?

Evaluation and evidence

- What evaluation design fits the question and how the data were collected (for example, a split, cross-validation, control, stability check, or data-quality diagnostic)?
- What metric or diagnostic will you use, and why does it match the question?
- How will you compare your work with a baseline or other reference? What would count as evidence against your initial expectation?
- What informative error, failure, uncertainty, tradeoff, or limitation will you inspect?
- What figures, tables, examples, or other outputs could make the results understandable?

Risks and backup scope

- What could prevent the plan from working (for example, access, collection, compute, time, data quality, or competition constraints)?
- What smaller version would still let you answer a useful question if a risk occurs?
- What evidence or feedback would lead you to pivot, and how would you document the reason?

Reproducibility and team contributions

- Where will code, configuration, data instructions, and results live in the repository?

- What is the first small runnable result you aim to produce, and what will a reader need to reproduce it?
- How will team members contribute substantively and review one another's work?
- What support, approval, or early instructor feedback would help you proceed?

Sources

List initial data sources, papers, competition pages, public notebooks, software, and other materials relevant to the plan. Include links and access or license details when known.

- [Source or material, link, and relevant note]